4. Scenario: You are working on a project that involves analyzing the sales performance of a company over the past four quarters. The quarterly sales data is stored in a NumPy array named sales_data, where each element represents the sales amount for a specific quarter. Your task is to calculate the total sales for the year and determine the percentage increase in sales from the first quarter to the fourth quarter. Question: Using NumPy arrays and arithmetic operations calculate the total sales for the year and determine the percentage increase in sales from the first quarter to the fourth quarter?

AIM:

To calculate the total annual sales and percentage increase in sales from the first quarter to the fourth quarter using NumPy arrays and arithmetic operations.

PROCEDURE:

1. Import the NumPy library.
2. Create a NumPy array containing quarterly sales data.
3. Calculate total annual sales using np.sum().
4. Calculate the percentage increase from the first quarter to the fourth quarter.
5. Display the total sales and percentage increase.

PROGRAM CODE:

```
import numpy as np

sales_data = np.array([10000, 12000, 15000, 18000])

total_sales = np.sum(sales_data)

increase = ((sales_data[3] - sales_data[0]) / sales_data[0]) * 100
```

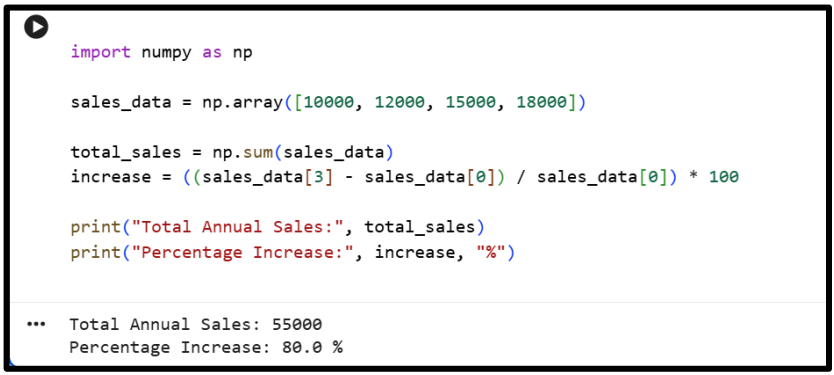
print("Total Annual Sales:", total_sales)

print("Percentage Increase:", increase, "%")

INPUT:

Quarterly sales = [10000, 12000, 15000, 18000]

OUTPUT:



```
import numpy as np

sales_data = np.array([10000, 12000, 15000, 18000])

total_sales = np.sum(sales_data)
increase = ((sales_data[3] - sales_data[0]) / sales_data[0]) * 100

print("Total Annual Sales:", total_sales)
print("Percentage Increase:", increase, "%")
```

```
Total Annual Sales: 55000
Percentage Increase: 80.0 %
```

RESULT:

Thus, the total annual sales are 55,000, and the percentage increase in sales from the first quarter to the fourth quarter is 80%.